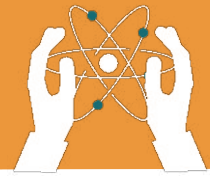


UNIT 6

ELECTRIC CURRENT AND CIRCUITS

Learning objectives:

- Understand how electric current flows through a complete circuit.
- Identify the components of an electric circuit, such as the power source, conductors, load, and switch.
- Recognize the importance of safety precautions when using electricity to avoid accidents.



What happens if we switch off the light in room?

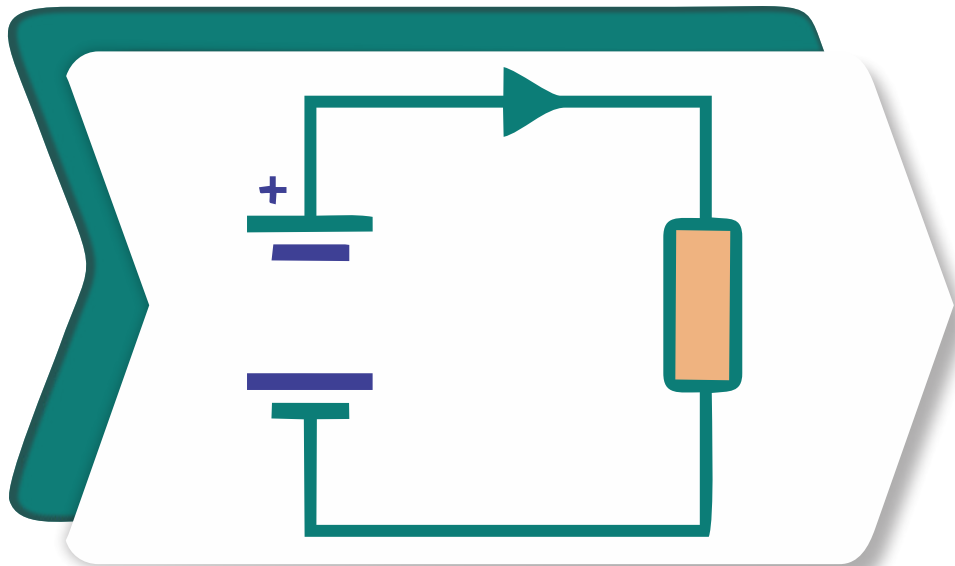
Electricity powers our daily lives, from lighting our homes to running appliances. To understand how it works, this chapter explains electric circuits, their components, and the difference between open and closed circuits.

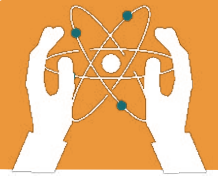
What is Electric current?

Electric current is the flow of electricity through a circuit, much like water flows through a pipe. This flow of electric energy is what powers devices such as bulbs, fans, and televisions.

How Does Electric Current Flow?

Electric current flows through a complete path called a circuit. The components attached in a circuit enable electricity to flow.





What Are Electric Circuits?

An **electric circuit** is a complete path or channel that allows electric current to flow. It is made up of several components that work together to power devices.

Conductors and Insulators

1. Conductors: Materials that allow electricity to pass through them, like metals (e.g., copper and aluminum).

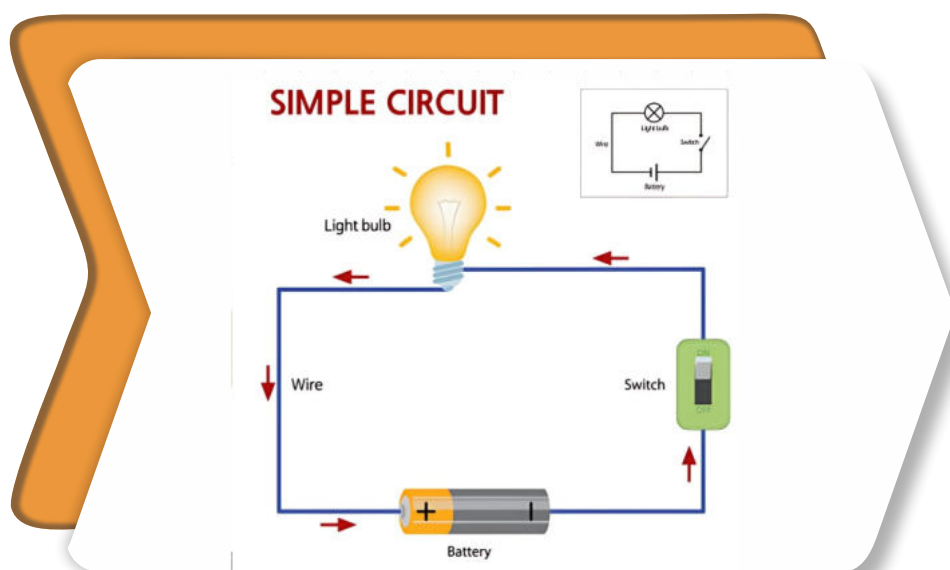
2. Insulators: Materials that do not allow electricity to pass through, like rubber, plastic, and wood.

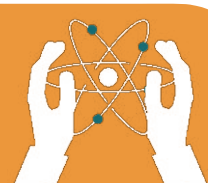
Components of Electric Circuits

1. Power Source: Supplies the energy needed to make electricity flow. Examples: Batteries or generators.

2. Conductors: Wires that carry the electric current from one part of the circuit to another.

3. Load: A device that uses electricity, such as a bulb, fan, or motor.





Open and Closed Circuits

1. Closed Circuit:

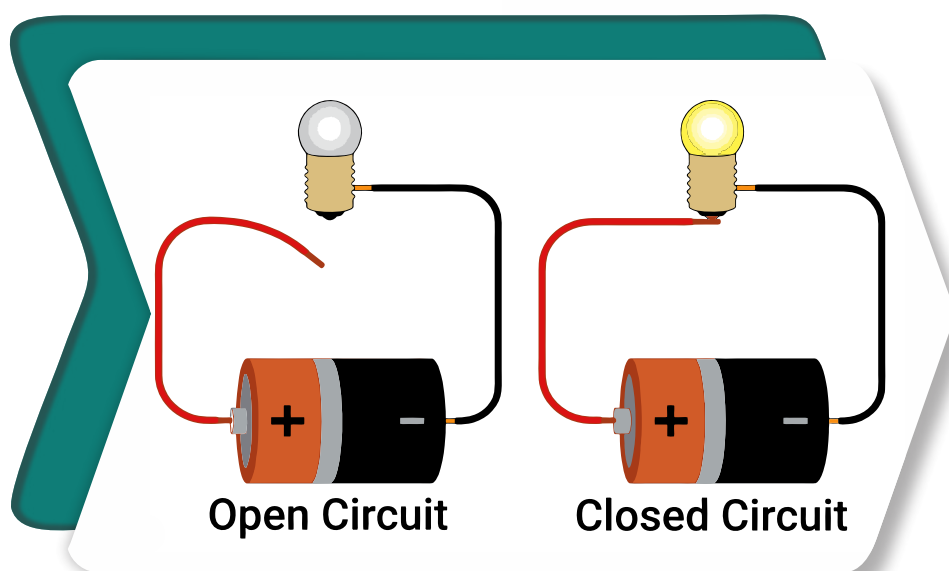
A closed circuit is a complete path that allows electricity to flow. When the switch is ON, the circuit is closed, and the device works.

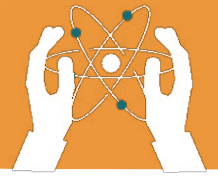
Example: A light bulb glows when we turn the switch on.

2. Open Circuit:

An open circuit is an incomplete path, so electricity cannot flow. When the switch is OFF, the circuit is open, and the device stops working.

Example: A bulb that does not glow when the switch is turned OFF.





Types of Circuits

1. Series Circuit:

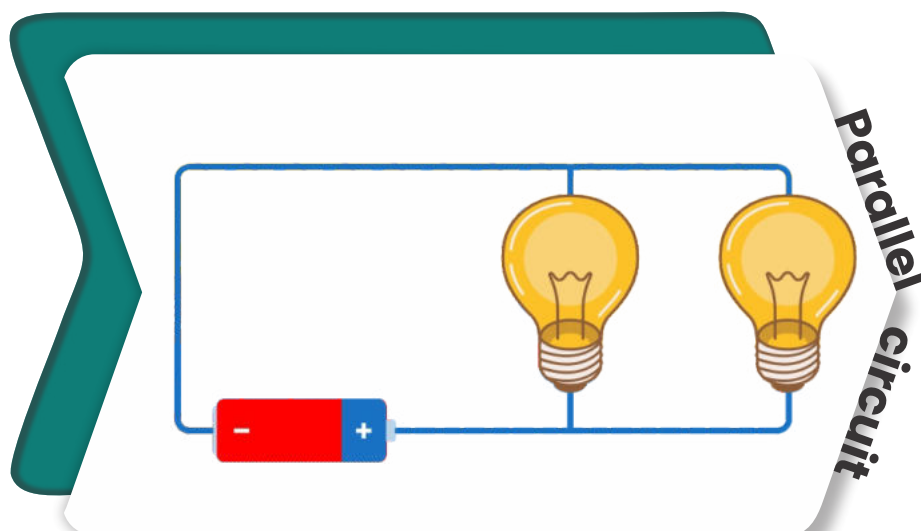
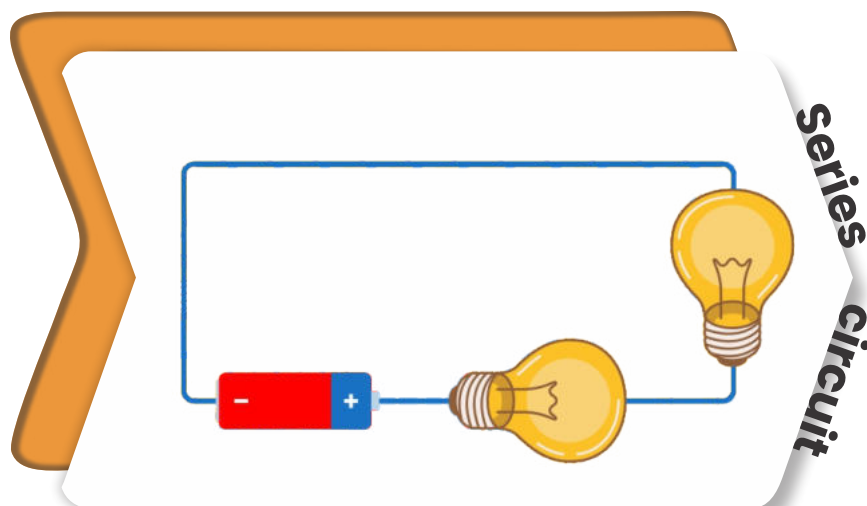
- Components are connected one after another in a single path.
- If one-part stops working, the entire circuit stops.

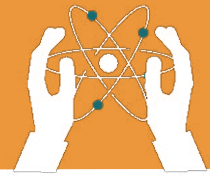
Example: Old-fashioned Christmas lights.

2. Parallel Circuit:

- Components are connected side by side.
- If one-part stops working, the other parts continue to work.

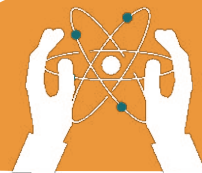
Example: The wiring in your home.





Safety Tips When Using Electricity

1. Never touch electrical devices or switches with wet hands.
2. Avoid using damaged or exposed wires.
3. Always turn off appliances when not in use.
4. Use insulated tools when handling electrical devices.
5. Stay away from power lines and never insert objects into sockets.

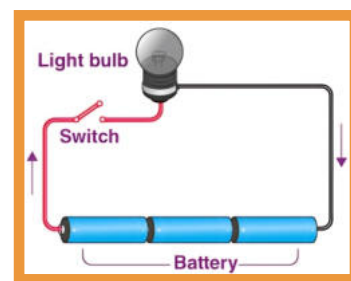


EXERCISES

(a) Some children make this circuit to light a bulb.

The bulb is not lit.

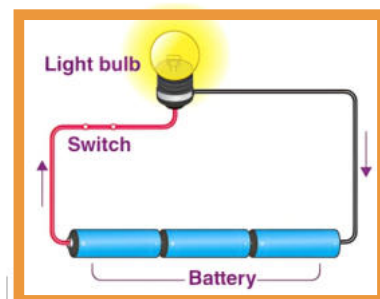
Why is the bulb not lit?

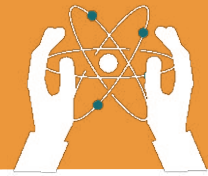


(b) The children make another circuit to light a bulb.

The bulb is lit.

Why is the bulb lit?



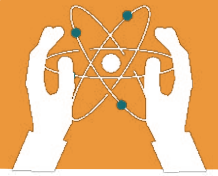


ELECTRIC CURRENT AND CIRCUITS

Fill in the blanks.

1. Electric current flows through a _____.
2. A _____ supplies energy to an electric circuit.
3. A _____ is used to control the flow of electricity.
4. A light bulb is an example of a _____ in a circuit.
5. An _____ circuit stops electricity from flowing.
6. _____ are materials that allow electricity to pass through them.
7. A _____ circuit allows devices to keep working even if one component stops.
8. Electric current is like _____ flowing through a pipe.

ELECTRIC CURRENT AND CIRCUITS



Write “T” for True or “F” for False.

1. A battery is a type of load.

2. An open circuit allows electricity to flow.

3. A switch is used to open or close a circuit.

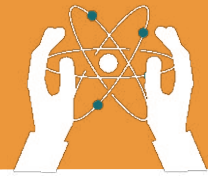
4. Wires are examples of insulators.

5. A parallel circuit stops working if one component fails.

6. A generator is a source of electricity.

7. Electricity flows through a complete circuit.

8. A series circuit has a single path for electricity.



Choose the correct answer.

1. Which component provides power to a circuit?

- Bulb
 - Battery
 - Switch
 - Wire
-

2. What is a closed circuit?

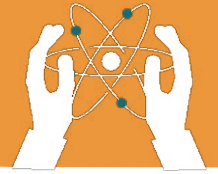
- A circuit with a complete path for electricity
 - A circuit with no electricity
 - A circuit with a broken wire
 - A circuit with no load
-

3. Which material is an insulator?

- Copper
 - Aluminum
 - Plastic
 - Iron
-

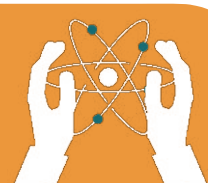
4. What happens in a parallel circuit?

- Electricity flows through a single path
- All devices stop working if one fails
- Electricity flows through multiple paths
- The circuit never works



5. Which is an example of a load?

- Battery
 - Light bulb
 - Wire
 - Switch
-



Answer the following questions.

Q1. What is an electric circuit?

Q2. Name the components of an electric circuit.

Q3. How are series circuits and parallel circuits different?

Q4. Why does a bulb not glow in an open circuit?

Q5. Make a list of materials that are conductors and insulators?

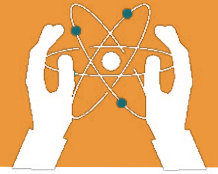
Q6. What role does a switch play in a circuit?



Critical Thinking

Why can electricity flow through water when they come into contact? What causes the electric current to travel through the water?

ELECTRIC CURRENT AND CIRCUITS



Build a Simple Electric Circuit



ACTIVITY

- **Materials Needed:** Battery, two wires, and small bulb.

Steps:

1. Connect one wire from the battery to the bulb.
2. Connect the other wire from the bulb back to the battery.
3. Observe the bulb glowing when the circuit is complete!